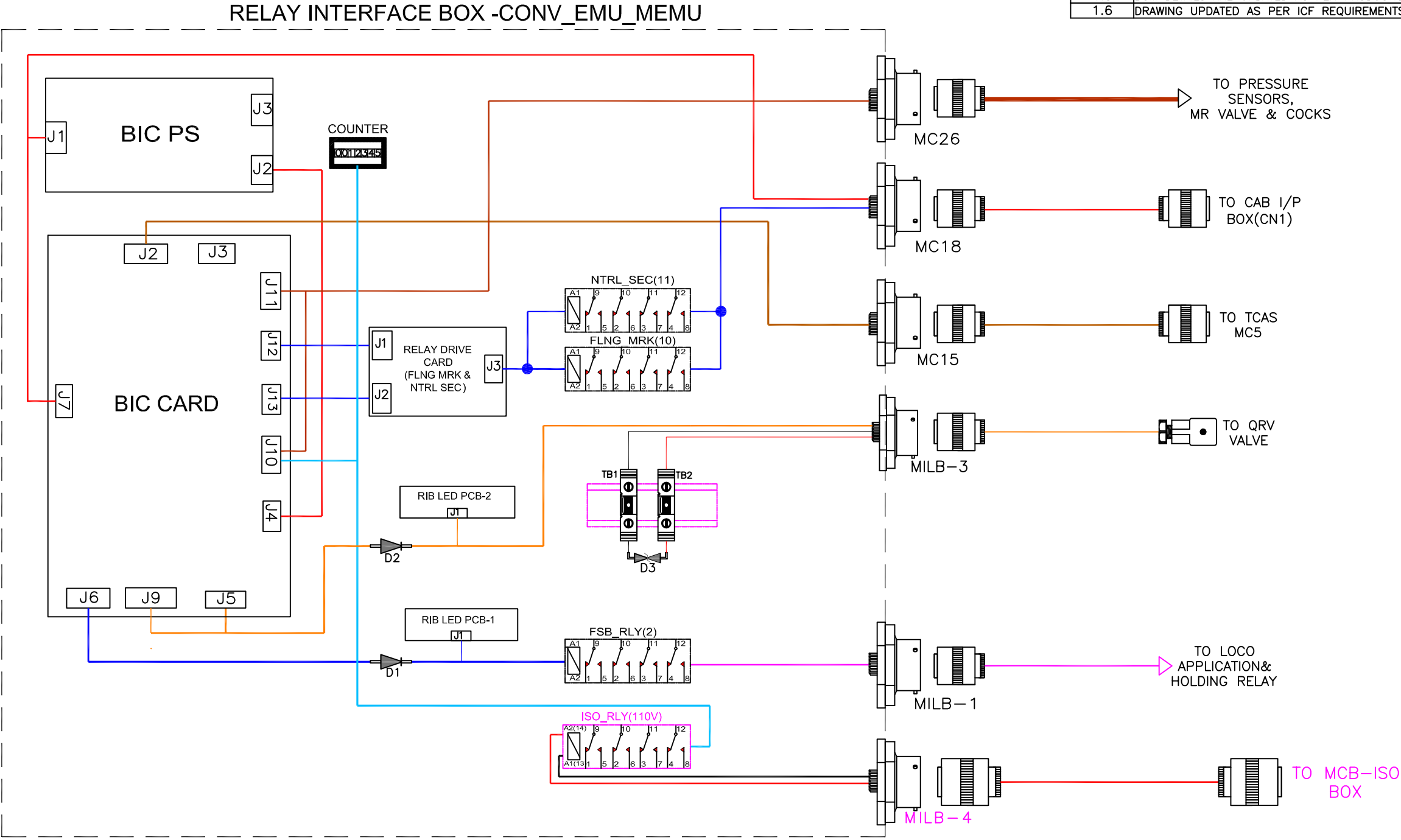


REV.NO.	CHANGE DETAILS	DATE	PREPARED	APPD.
1.1	ADDED FLNG MARK& NTRL SEC WIRING	06.11.24	DLR	YS
1.2	ISOLATION SW WIRING INTERCHANGED AS PER CoE	25.06.25	DLR	YS
1.3	WIRING UPDATED WITH TRACTION CUT-OFF FEEDBACK	01.07.25	DLR	YS
1.4	WIRING LENGTHS UPDATED AS PER PRODUCTION	26.07.25	DLR	YS
1.5	BOM DETAILS ARE UPDATED AS PER PRODUCTION	05.10.25	DLR	YS
1.6	DRAWING UPDATED AS PER ICF REQUIREMENTS	29.05.26	DLR	YS

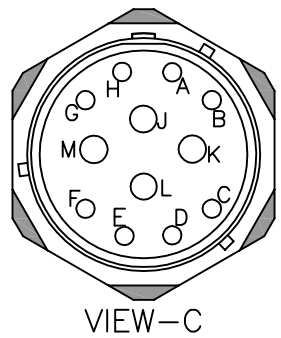


Note:-

1.Test the all assembled cables as per testing instruction. Ref. No: 5 16 90 008

PROJECT: KAVACH		HBL HBL Engineering Ltd.	NAME		DESIGNATION	SIGN.
ASSEMBLY: EMU—MEMU			DESIGN	D.Lovaraju	Dy.Manager	
			CHECKED	J.V.Prasad	Sr.Manager	
			APPROVED	Y.Subrahmanyam	GM(Engg.)	
SCALE: NTS	SIZE: A3	TITLE: HARNS_INTRNL_RIB_CONV_EMU_MEMU (RIB_BLOCK DIAGRAM)			DRAWING NO. 516490446	REV. 1.6
	SHEET: 1 OF 13				PART NO. 6000050624	
EFFECTIVE FROM:			TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 – 1993.			
ALL DIMENSIONS ARE IN MM			THIS DRAWING IS THE PROPERTY OF HBL ENGINEERING LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.			

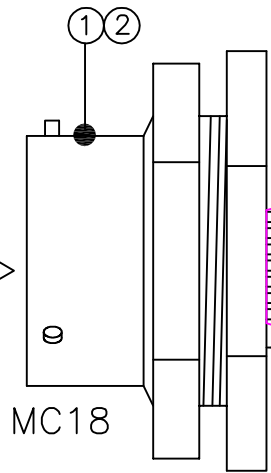
REV.NO.	CHANGE DETAILS	DATE	PREPARED	APPD.
1.1	ADDED FLNG MARK& NTRL SEC WIRING	06.11.24	DLR	YS
1.2	ISOLATION SW WIRING INTERCHANGED AS PER CoE	25.06.25	DLR	YS
1.3	WIRING UPDATED WITH TRACTION CUT-OFF FEEDBACK	01.07.25	DLR	YS
1.4	WIRING LENGTHS UPDATED AS PER PRODUCTION	26.07.25	DLR	YS
1.5	BOM DETAILS ARE UPDATED AS PER PRODUCTION	05.10.25	DLR	YS
1.6	DRAWING UPDATED AS PER ICF REQUIREMENTS	29.05.26	DLR	YS



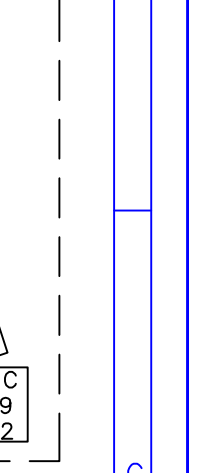
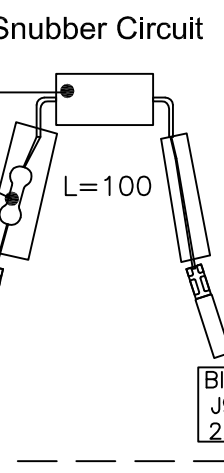
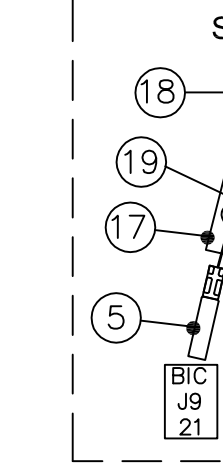
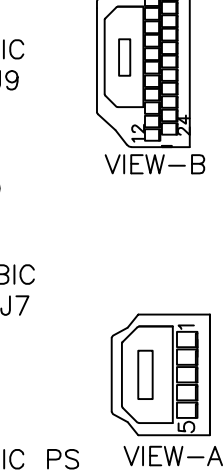
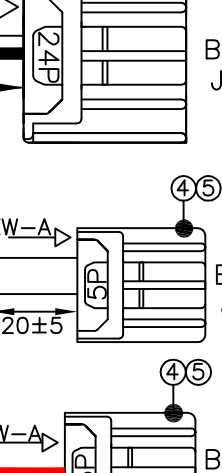
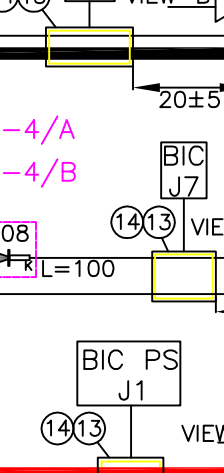
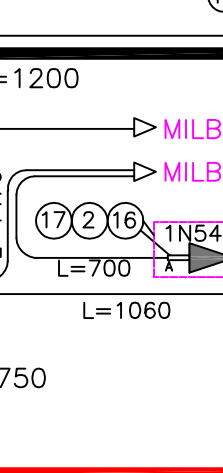
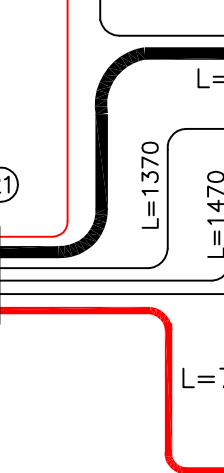
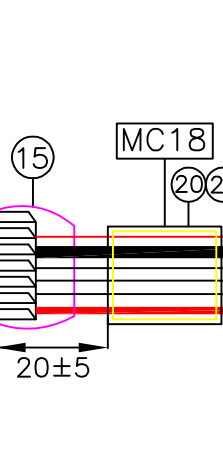
FRONT VIEW
12PIN
CONTACT
(MALE)

VIEW-C

VIEW-C

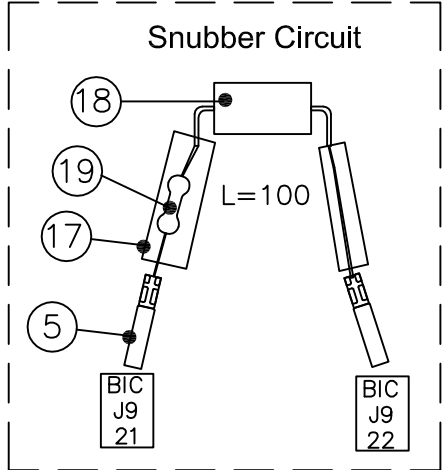


MC18



PART LIST				
ITEM NO/REF	QTY	UNITS	ITEM CODE	DESCRIPTION
1	1.00	PC	1000088346	CONNECTOR_RECEP_12P_62IN-5024-14-12P
2	0.20	MTR	1000016883	SLEEVE_HEAT SHINKABLE_DIA 3MM_BLACK
3	1.00	PC	1000025258	CONNECTOR_SOCKET_24POS_PNO:MX34024SF1
4	2.00	PC	1000027695	CONNECTOR_SOCKET_5POS_MX34005SF1
5	11.00	PC	1000025249	CONNECTOR_CRIMP_TERMINAL_MX34
6	1.00	PC	1000010068	LUG_1.5SQ MM_CU_RING TYPE M6
7	1.81	MTR	1000013442	PTFE_WIRE_AWG 22(7/30)_BLACK
8	3.81	MTR	1000025384	WIRE_PTFE_RED_AWG 22_MIL-16878E/4BFB-2
9	2.27	MTR	1000025386	WIRE_PTFE_YELLOW_AWG 22_MIL16878E/4BFB-4
10	1.00	MTR	1000025381	WIRE_PTFE_GREEN_AWG 22_MIL16878E/4BFB-5
11	1.20	MTR	1000025379	WIRE_PTFE_BLUE_AWG 22_MIL-16878E/4BFB-6
12	1.37	MTR	1000025382	WIRE_PTFE_GREY_AWG 22_MIL-16878E/4BFB-8
13	0.15	MTR	1000075057	HEAT_SHRINKABLE_SLEEVE_5MM_YELLOW
14	0.20	MTR	1000075058	HEAT_SHRINKABLE_SLEEVE_5MM_CLEAR
15	0.10	MTR	1000074458	HEAT-SHRINKABLE_SLEEVE_16MM_BLACK
16	1.00	PC	1000075399	DIODE_SCHOTTKY_LEADED_40V/5A
17	0.10	MTR	1000026054	SLEEVE_HEAT SHRINKABLE_5MM_BLACK
18	1.00	PC	1000003441	CAPACITOR_MPEAR(AXIAL)_0.47MFD/250V
19	1.00	PC	1000014018	RESISTOR_MFR_47E_1/4W_1%
20	0.05	MTR	1000016947	SLEEVE_MARKER_3/8"_YELLOW
21	0.06	MTR	1000016866	SLEEVE_CLEAR_3/8"

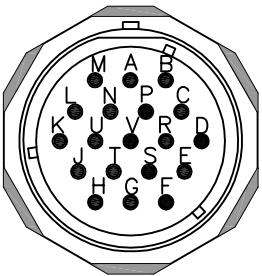
WIRE TERMINATION								
MC18	BIC PS J1	BIC J9	BIC J7	MILB-4	SIGNAL	WIRE COLOUR	BOM REF#	LENGTH
J	1	-	-	-	110V_+VE	RED	8	750
L	2	-	-	-	110V_+VE	RED	8	
K	4	-	-	-	110_GND	BLACK	7	
M	5	-	5	-	110_GND	BLACK	7	1060
G	-	-	-	D	LOCO_110V	RED	8	1410
-	-	9	-	E	REG_COM	RED	8	900
H	-	8	-	-	REG_COIL	BLUE	11	1200
E	-	-	-	A	1412A	GREY	12	1370
F	-	-	-	B	MCC_1	YELLOW	9	1470
-	-	-	3	B	MCC_1	YELLOW	9	700+100
-	3	-	-	-	CHASSIS	GREEN	10	1000



Note:-
1.Label the all connectors as visible at respective location.
2.Add Diode(1N5408) inseries with the wires and add heatshrinkable sleeve, as shown in drawing.

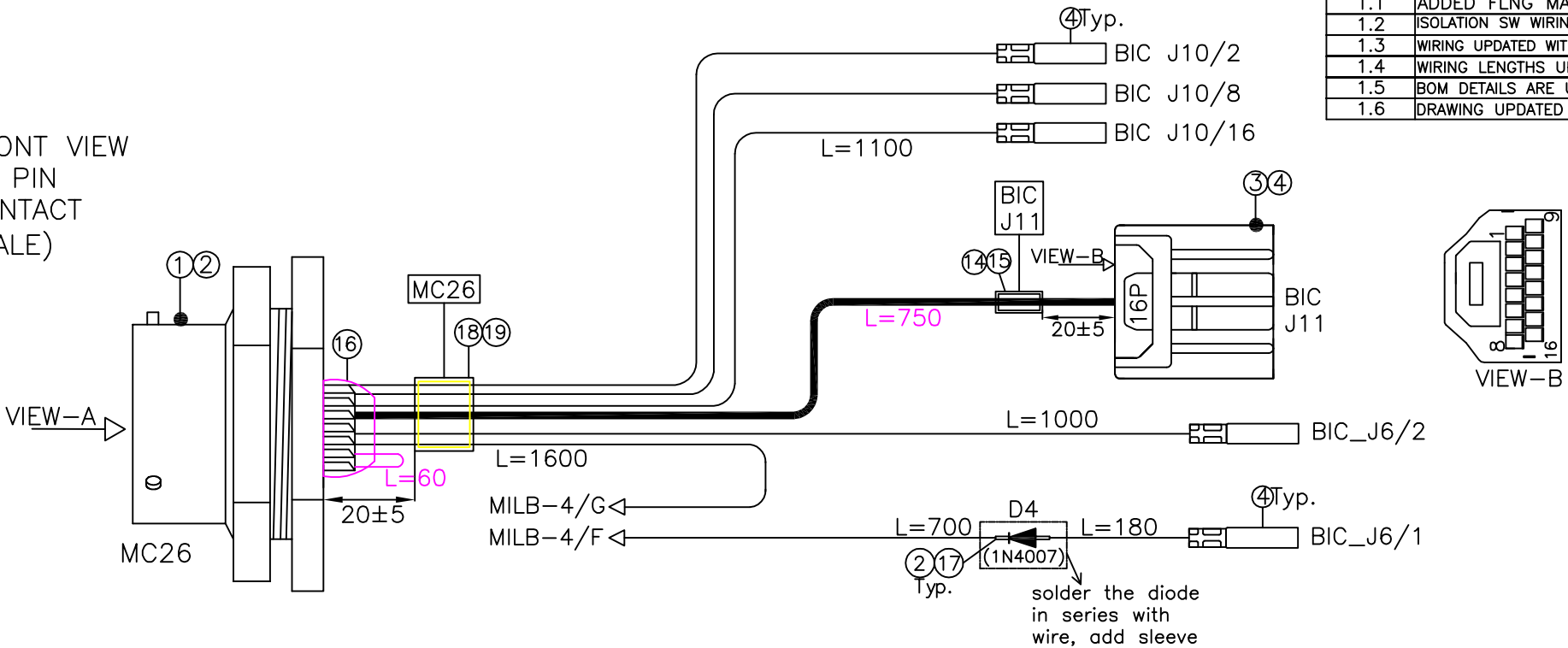
PROJECT: KAVACH	HBL HBL Engineering Ltd.	NAME	DESIGNATION	SIGN.
ASSEMBLY: EMU-MEMU		DESIGN	D.Lovaraju	Dy.Manager
		CHECKED	J.V.Prasad	Sr.Manager
SCALE: NTS	SIZE: A3	APPROVED	Y.Subrahmanyam	GM(Engg.)
TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 - 1993.		THIS DRAWING IS THE PROPERTY OF HBL ENGINEERING LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.		
EFFECTIVE FROM:		ALL DIMENSIONS ARE IN MM		
DRAWING NO. 516490446		REV. 1.6		
PART NO. 6000050624				

REV.NO.	CHANGE DETAILS	DATE	PREPARED	APPD.
1.1	ADDED FLNG MARK& NTRL SEC WIRING	06.11.24	DLR	YS
1.2	ISOLATION SW WIRING INTERCHANGED AS PER CoE	25.06.25	DLR	YS
1.3	WIRING UPDATED WITH TRACTION CUT-OFF FEEDBACK	01.07.25	DLR	YS
1.4	WIRING LENGTHS UPDATED AS PER PRODUCTION	26.07.25	DLR	YS
1.5	BOM DETAILS ARE UPDATED AS PER PRODUCTION	05.10.25	DLR	YS
1.6	DRAWING UPDATED AS PER ICF REQUIREMENTS	29.05.26	DLR	YS



FRONT VIEW
19 PIN
CONTACT
(MALE)

VIEW-A



PART LIST

ITEM NO/REF	QTY	UNITS	ITEM CODE	DESCRIPTION
1	1.00	PC	1000050114	CONNECTOR_PLUG_19PIN_62IN-5024-14-19P
2	0.30	MTR	1000016883	SLEEVE_HEAT SHINKABLE_DIA 3MM_BLACK
3	1.00	PC	1000025256	CONNECTOR_SOCKET_16POS_PNO:MX34016SF1
4	12.00	PC	1000025249	CONNECTOR_CRIMP_TERMINAL_MX34
5	2.20	MTR	1000013442	PTFE_WIRE_AWG 22(7/30)_BLACK
6	0.75	MTR	1000025380	WIRE_PTFE_BROWN_AWG 22_MIL-16878E/4BFB-1
7	0.75	MTR	1000025383	WIRE_PTFE_ORANGE_AWG22_MIL-16878E/4BFB-3
8	0.75	MTR	1000025386	WIRE_PTFE_YELLOW_AWG 22_MIL16878E/4BFB-4
9	1.10	MTR	1000025381	WIRE_PTFE_GREEN_AWG 22_MIL16878E/4BFB-5
10	3.23	MTR	1000025379	WIRE_PTFE_BLUE_AWG 22_MIL-16878E/4BFB-6
11	0.75	MTR	1000025382	WIRE_PTFE_GREY_AWG 22_MIL-16878E/4BFB-8
12	0.75	MTR	1000026453	PTFE_WIRE_AWG 22(7/30)_WHITE
13	1.10	MTR	1000025384	WIRE_PTFE_RED_AWG 22_MIL-16878E/4BFB-2
14	0.10	MTR	1000075057	HEAT_SHRINKABLE_SLEEVE_5MM_YELLOW
15	0.15	MTR	1000075058	HEAT_SHRINKABLE_SLEEVE_5MM_CLEAR
16	0.06	MTR	1000074458	HEAT-SHRINKABLE_SLEEVE_16MM_BLACK
17	1.00	PC	1000006449	DIODE_LEADED_1N4007_DO-41
18	0.05	MTR	1000016947	SLEEVE_MARKER_3/8"_YELLOW
19	0.06	MTR	1000016866	SLEEVE_CLEAR_3/8"

WIRE TERMINATION

MC26	BIC J11	BIC J10	BIC J6	MILB-4	SIGNAL	WIRE COLOUR	BOM REF#	LENGTH
M	5	-	-	-	BC_SENSOR +VE	BROWN	6	750
N	13	-	-	-	BC_PR RETURN	WHITE	12	
A	1	-	-	-	BP_SENSOR +VE	GREY	11	
B	9	-	-	-	BP_PR RETURN	BLUE	10	
K	4	-	-	-	MR_SENSOR +VE	YELLOW	8	
L	12	-	-	-	MR_PR RETURN	ORANGE	7	
F	-	8	-	-	BP ISOLATION COCK	GREEN	9	1100
S	-	2	-	-	24V_GND	BLACK	5	
D	-	16	-	-	MR ISOLATION COCK	RED	13	1100
E	-	2	-	-	24V_GND	BLACK	5	60(S&E loop)
-	-	-	1	F	MR_SOL_VALVE	BLUE	10	700+180
U	-	-	-	G	MR_SOL_VALVE	BLUE	10	1600
V	-	-	2	-	24V_GND(MR_VALVE)	BLACK	5	1000

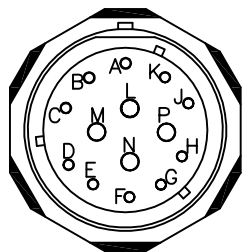
*LOOP S&E PINS ON MC26

Note:-

1. Label the all connectors as visible at respective location
2. Add required ferrules to cables which are connected to isolation switch at respective location.

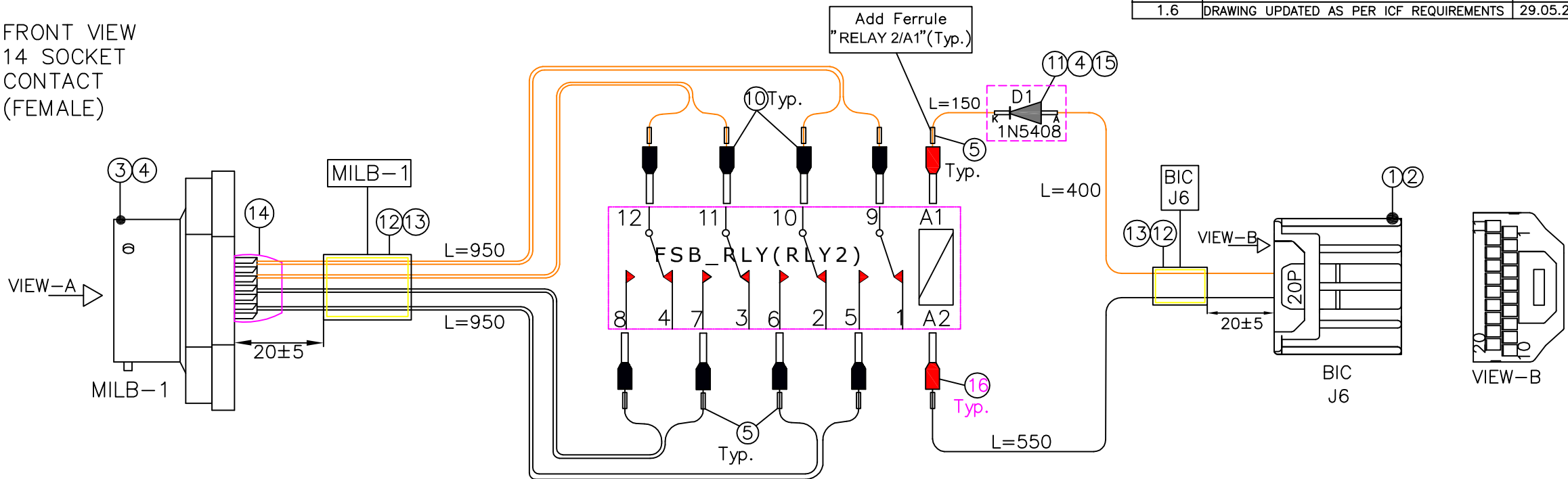
PROJECT: KAVACH	HBL HBL Engineering Ltd.	NAME	DESIGNATION	SIGN.
ASSEMBLY: EMU-MEMU		DESIGN	D.Lovaraju	Dy.Manager
		CHECKED	J.V.Prasad	Sr.Manager
SCALE: NTS	SIZE: A3	TITLE: HARNS_INTRNL_RIB_CONV_EMU_MEMU	DRAWING NO. 516490446	REV. 1.6
	SHEET: 4 OF 13	HARNS ASY_PRESSURE SENSORS & COCKS	PART NO. 6000050624	
EFFECTIVE FROM:		TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 - 1993.		
ALL DIMENSIONS ARE IN MM		THIS DRAWING IS THE PROPERTY OF HBL ENGINEERING LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.		

REV.NO.	CHANGE DETAILS	DATE	PREPARED	APPD.
1.1	ADDED FLNG MARK& NTRL SEC WIRING	06.11.24	DLR	YS
1.2	ISOLATION SW WIRING INTERCHANGED AS PER CoE	25.06.25	DLR	YS
1.3	WIRING UPDATED WITH TRACTION CUT-OFF FEEDBACK	01.07.25	DLR	YS
1.4	WIRING LENGTHS UPDATED AS PER PRODUCTION	26.07.25	DLR	YS
1.5	BOM DETAILS ARE UPDATED AS PER PRODUCTION	05.10.25	DLR	YS
1.6	DRAWING UPDATED AS PER ICF REQUIREMENTS	29.05.26	DLR	YS



FRONT VIEW
14 SOCKET
CONTACT
(FEMALE)

VIEW-A



PART LIST

ITEM NO/REF	QTY	UNITS	ITEM CODE	DESCRIPTION
1	1.00	PC	1000025257	CONNECTOR_SOCKET_20POS_PNO:MX34020SF1
2	2.00	PC	1000025249	CONNECTOR_CRIMP_TERMINAL_MX34
3	1.00	PC	1000088343	CONNECTOR_RECEP_14P_62IN-5024-14-A14-S
4	0.10	MTR	1000016883	SLEEVE_HEAT SHINKABLE_DIA 3MM_BLACK
5	0.30	MTR	1000007218	FERRULE_TUBE_3.3MM_WHITE
6	0.55	MTR	1000013442	PTFE_WIRE_AWG 22(7/30)_BLACK
7	0.55	MTR	1000025383	WIRE_PTFE_ORANGE_AWG22_MIL-16878E/4BFB-3
8	3.80	MTR	1000025386	WIRE_PTFE_YELLOW_AWG 22_MIL16878E/4BFB-4
9	3.80	MTR	1000025382	WIRE_PTFE_GREY_AWG 22_MIL-16878E/4BFB-8
10	8.00	PC	1000095964	FERRULE_INSULATED_0.5MM_58-508L8_JAINSON
11	1.00	PC	1000075399	DIODE_SCHOTTKY_LEADED_40V_5A
12	0.10	MTR	1000075057	HEAT_SHRINKABLE_SLEEVE_5MM_YELLOW
13	0.15	MTR	1000075058	HEAT_SHRINKABLE_SLEEVE_5MM_CLEAR
14	0.06	MTR	1000074458	HEAT_SHRINKABLE_SLEEVE_16MM_BLACK
15	0.10	MTR	1000026054	SLEEVE_HEAT SHRINKABLE_5MM_BLACK
16	2.00	PC	1000095965	FERRULE_INSULATED_1MM_58-510L8_JAINSON

Note:-

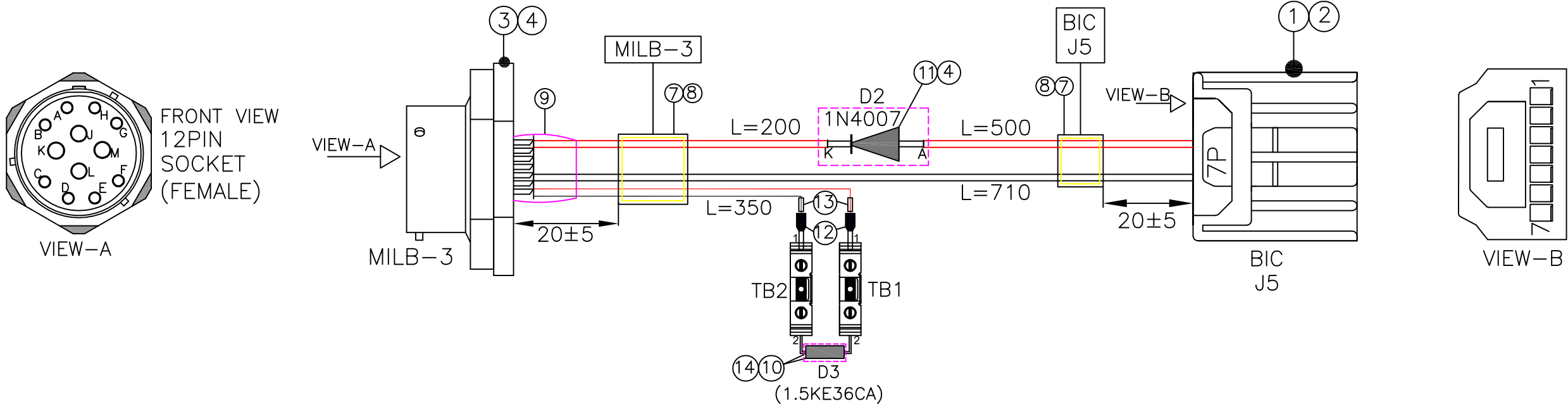
- Label the all connectors as visible at respective location
- Add required ferrules to cables which are connected to relays at respective location.
- Add sleeve to diode "D1"and shrink before soldering.

WIRE TERMINATION

BIC J6	FSB RLY	MILB-1	SIGNAL	WIRE COLOUR	BOM REF#	LENGTH
3	A1	-	FSB_RLY_COIL	ORANGE	7	400+150
4	A2	-	GND_24V	BLACK	6	550
-	9&10	M	FSB_RLY_COM1	YELLOW	8	950&950
-	11&12	L	FSB_RLY_COM2	YELLOW	8	950&950
-	5&6	N	FSB_RLY_NO1	GREY	9	950&950
-	7&8	P	FSB_RLY_NO2	GREY	9	950&950

PROJECT: KAVACH		HBL HBL Engineering Ltd.		NAME	DESIGNATION	SIGN.
ASSEMBLY: EMU—MEMU			DESIGN	D.Lovaraju	Dy.Manager	
			CHECKED	J.V.Prasad	Sr.Manager	
			APPROVED	Y.Subrahmanyam	GM(Engg.)	
SCALE: NTS	SIZE: A3	TITLE: HARNS_INTRNL_RIB_CONV_EMU_MEMU HARNS ASY_RELAY DRIVE SIGNALS			DRAWING NO. 516490446	REV. 1.6
	SHEET: 5 OF 13				PART NO. 6000050624	
EFFECTIVE FROM: ALL DIMENSIONS ARE IN MM			TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 – 1993. THIS DRAWING IS THE PROPERTY OF HBL ENGINEERING LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.			

REV.NO.	CHANGE DETAILS	DATE	PREPARED	APPD.
1.1	ADDED FLNG MARK& NTRL SEC WIRING	06.11.24	DLR	YS
1.2	ISOLATION SW WIRING INTERCHANGED AS PER CoE	25.06.25	DLR	YS
1.3	WIRING UPDATED WITH TRACTION CUT-OFF FEEDBACK	01.07.25	DLR	YS
1.4	WIRING LENGTHS UPDATED AS PER PRODUCTION	26.07.25	DLR	YS
1.5	BOM DETAILS ARE UPDATED AS PER PRODUCTION	05.10.25	DLR	YS
1.6	DRAWING UPDATED AS PER ICF REQUIREMENTS	29.05.26	DLR	YS



PART LIST				
ITEM NO/REF	QTY	UNITS	ITEM CODE	DESCRIPTION
1	1.00	PC	1000041208	CONNECTOR_SOCKET_7POS_MX34007SF1
2	4.00	PC	1000025249	CONNECTOR_CRIMP_TERMINAL_MX34
3	1.00	PC	1000088347	CONNECTOR_RECEP_12P_62IN-5024-14-12S
4	0.20	MTR	1000016883	SLEEVE_HEAT SHRINKABLE_DIA 3MM_BLACK
5	1.06	MTR	1000013442	PTFE_WIRE_AWG 22(7/30)_BLACK
6	1.77	MTR	1000025384	WIRE_PTFE_RED_AWG 22_MIL-16878E/4BFB-2
7	0.10	MTR	1000075057	HEAT_SHRINKABLE_SLEEVE_5MM_YELLOW
8	0.12	MTR	1000075058	HEAT_SHRINKABLE_SLEEVE_5MM_CLEAR
9	0.10	MTR	1000074458	HEAT-SHRINKABLE_SLEEVE_16MM_BLACK
10	1.00	PC	1000024955	DIODE_LEADED_1.5KE36CA_36V/1500W
11	1.00	PC	1000006449	DIODE_LEADED_1N4007_DO-41
12	2.00	PC	1000095964	FERRULE_INSULATED_0.5MM_58-508L8_JAINSON
13	0.10	MTR	1000007218	FERRULE_TUBE_3.3MM_WHITE
14	0.10	MTR	1000026054	SLEEVE_HEAT SHRINKABLE_5MM_BLACK

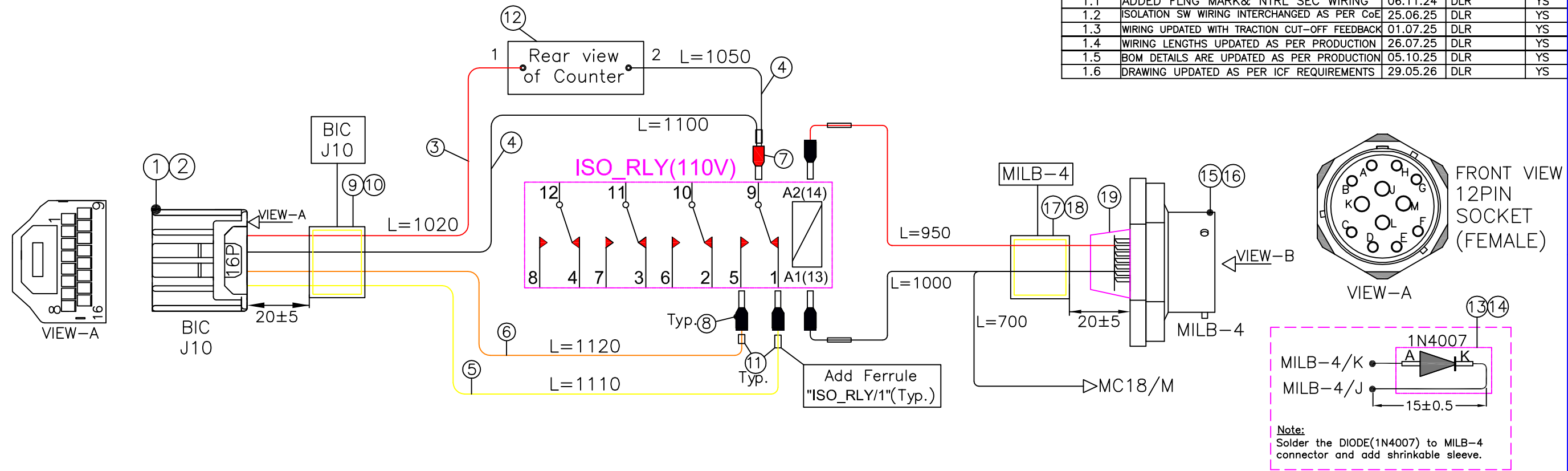
WIRE TERMINATION								
MILB-3	DIODE (D2)	BIC J5	TB1	TB2	SIGNAL	WIRE COLOUR	BOM REF#	LENGTH
J,K	K	—	—	—	QRV_COIL_HIGH	RED	6	500+200
—	A	6,7	—	—	QRV_COIL_HIGH	RED	6	
A,B	—	3,4	—	—	QRV_COIL_LOW	BLACK	5	710
J	—	—	1	—	QRV_COIL_HIGH	RED	6	350
A	—	—	—	1	QRV_COIL_LOW	BLACK	5	

Note:—

- Label the all connectors as visible at respective location
- Add required ferrules to cables which are connected to relays at respective location.
- Solder the diode "D2" in series with wires and add shrinkable sleeve and shrink.

PROJECT: KAVACH		HBL HBL Engineering Ltd.		NAME	DESIGNATION	SIGN.	
ASSEMBLY: EMU—MEMU			DESIGN	D.Lovaraju	Dy.Manager		
			CHECKED	J.V.Prasad	Sr.Manager		
			APPROVED	Y.Subrahmanyam	GM(Engg.)		
SCALE: NTS	SIZE: A3	TITLE: HARNS_INTRNL_RIB_CONV_EMU_MEMU			DRAWING NO. 516490446		REV. 1.6
		HARNS ASY_QRV SOLENOID VALVE			PART NO. 6000050624		
EFFECTIVE FROM: ALL DIMENSIONS ARE IN MM				TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 — 1993. THIS DRAWING IS THE PROPERTY OF HBL ENGINEERING LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.			

REV.NO.	CHANGE DETAILS	DATE	PREPARED	APPD.
1.1	ADDED FLNG MARK& NTRL SEC WIRING	06.11.24	DLR	YS
1.2	ISOLATION SW WIRING INTERCHANGED AS PER CoE	25.06.25	DLR	YS
1.3	WIRING UPDATED WITH TRACTION CUT-OFF FEEDBACK	01.07.25	DLR	YS
1.4	WIRING LENGTHS UPDATED AS PER PRODUCTION	26.07.25	DLR	YS
1.5	BOM DETAILS ARE UPDATED AS PER PRODUCTION	05.10.25	DLR	YS
1.6	DRAWING UPDATED AS PER ICF REQUIREMENTS	29.05.26	DLR	YS



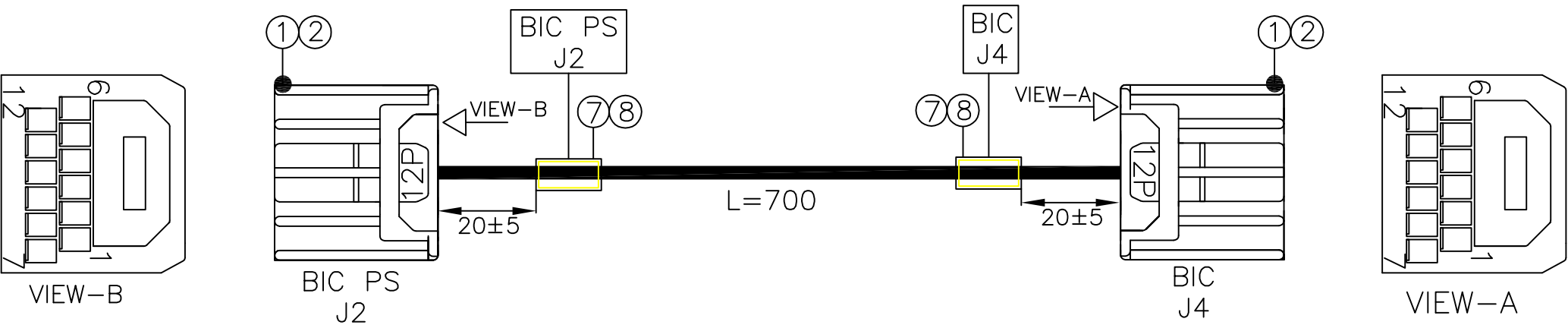
PART LIST				
ITEM NO/REF	QTY	UNITS	ITEM CODE	DESCRIPTION
1	1.00	PC	1000025256	CONNECTOR_SOCKET_16POS_PNO:MX34016SF1
2	4.00	PC	1000025249	CONNECTOR_CRIMP_TERMINAL_MX34
3	1.80	MTR	1000025384	WIRE_PTFE_RED_AWG 22_MIL-16878E/4BFB-2
4	4.00	MTR	1000013442	PTFE_WIRE_AWG 22(7/30)_BLACK
5	1.12	MTR	1000025383	WIRE_PTFE_ORANGE_AWG22_MIL-16878E/4BFB-3
6	1.11	MTR	1000025386	WIRE_PTFE_YELLOW_AWG 22_MIL16878E/4BFB-4
7	2.00	PC	1000095965	FERRULE_INSULATED_1MM_58-510L8_JAINSON
8	3.00	PC	1000095964	FERRULE_INSULATED_0.5MM_58-508L8_JAINSON
9	0.05	MTR	1000075057	HEAT_SHRINKABLE_SLEEVE_5MM_YELLOW
10	0.06	MTR	1000075058	HEAT_SHRINKABLE_SLEEVE_5MM_CLEAR
11	0.20	MTR	1000007218	FERRULE_TUBE_3.3MM_WHITE
12	1.00	PC	1000027376	EM_COUNTER_7_DIGIT_24V_DC_KUBLER
13	1.00	PC	1000006449	DIODE_LEADED_1N4007_DO-41
14	0.10	MTR	1000026054	SLEEVE_HEAT_SHRINKABLE_5MM_BLACK
15	1.00	PC	1000088347	CONNECTOR_RECEP_12P_62IN-5024-14-12S
16	0.20	MTR	1000016883	SLEEVE_HEAT_SHINKABLE_DIA 3MM_BLACK
17	0.05	MTR	1000016947	SLEEVE_MARKER_3/8"_YELLOW
18	0.06	MTR	1000016866	SLEEVE_CLEAR_3/8"
19	0.05	MTR	1000074458	HEAT-SHRINKABLE_SLEEVE_16MM_BLACK

WIRE TERMINATION								
BIC J10	ISO RELAY	COUNTER	MILB-4	MC18	SIGNAL	WIRE COLOUR	BOM REF#	LENGTH
14	5	-	-	-	ISO_SW_NO1	ORANGE	5	1120
13	1	-	-	-	ISO_SW_NC1	YELLOW	6	1110
11	-	1	-	-	COUNTER COIL	RED	3	1020
10	9	2	-	-	GND_24V	BLACK	4	1100+1050
-	A2(14)	-	J	-	110V_DC(ISO)	RED	3	950
-	A1(13)	-	K	M	110V_GND	BLACK	4	1000+700

Note:-
1. Label the all connectors as per respective location.
2. Counter wires will be soldered at the time of unit assembly.

PROJECT: KAVACH		HBL HBL Engineering Ltd.		NAME	DESIGNATION	SIGN.
ASSEMBLY: EMU—MEMU			DESIGN	D.Lovaraju	Dy.Manager	
			CHECKED	J.V.Prasad	Sr.Manager	
			APPROVED	Y.Subrahmanyam	GM(Engg.)	
SCALE: NTS	SIZE: A3	TITLE: HARNS_INTRNL_RIB_CONV_EMU_MEMU HARNS_ASSY_ISO SWITCH & COUNTER	DRAWING NO. 516490446		REV. 1.6	
	SHEET: 7 OF 13		PART NO. 6000050624			
EFFECTIVE FROM: ALL DIMENSIONS ARE IN MM			TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 – 1993. THIS DRAWING IS THE PROPERTY OF HBL ENGINEERING LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.			

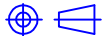
REV.NO.	CHANGE DETAILS	DATE	PREPARED	APPD.
1.1	ADDED FLNG MARK& NTRL SEC WIRING	06.11.24	DLR	YS
1.2	ISOLATION SW WIRING INTERCHANGED AS PER CoE	25.06.25	DLR	YS
1.3	WIRING UPDATED WITH TRACTION CUT-OFF FEEDBACK	01.07.25	DLR	YS
1.4	WIRING LENGTHS UPDATED AS PER PRODUCTION	26.07.25	DLR	YS
1.5	BOM DETAILS ARE UPDATED AS PER PRODUCTION	05.10.25	DLR	YS
1.6	DRAWING UPDATED AS PER ICF REQUIREMENTS	29.05.26	DLR	YS



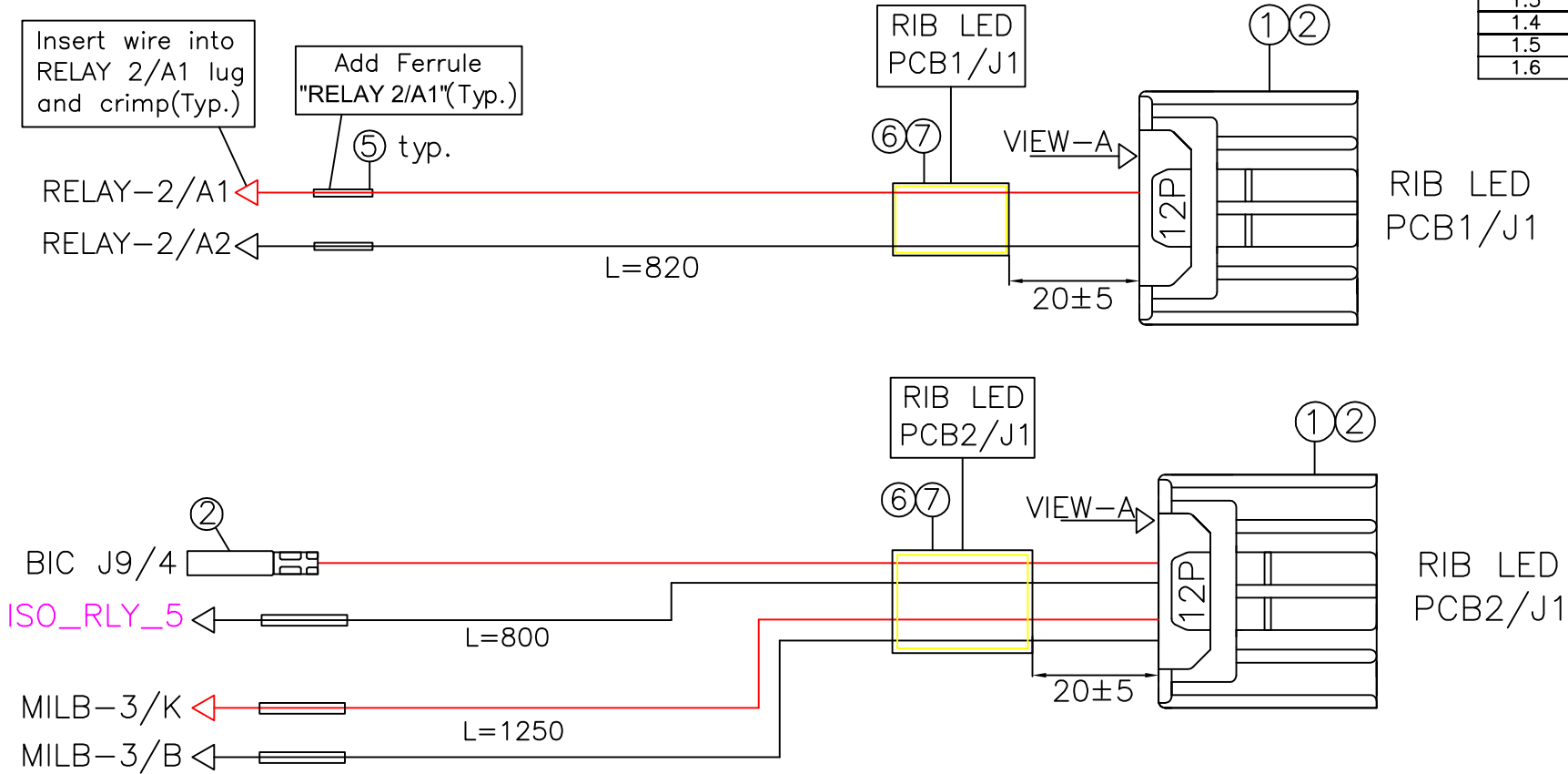
PART LIST				
ITEM NO/REF	QTY	UNITS	ITEM CODE	DESCRIPTION
1	2.00	PC	1000027692	CONNECTOR_SOCKET_12POS_MX34012SF1
2	20.00	PC	1000025249	CONNECTOR_CRIMP_TERMINAL_MX34
3	2.80	MTR	1000025384	WIRE_PTFE_RED_AWG 22_MIL-16878E/4BFB-2
4	2.80	MTR	1000013442	PTFE_WIRE_AWG 22(7/30)_BLACK
5	0.70	MTR	1000025379	WIRE_PTFE_BLUE_AWG 22_MIL-16878E/4BFB-6
6	0.70	MTR	1000025381	WIRE_PTFE_GREEN_AWG 22_MIL16878E/4BFB-5
7	0.10	MTR	1000016947	SLEEVE_MARKER_3/8"_YELLOW
8	0.12	MTR	1000016866	SLEEVE_CLEAR_3/8"

WIRE TERMINATION					
BIC PS J2	BIC J4	SIGNAL	WIRE COLOUR	BOM REF#	LENGTH
1	1	24V_ISO	RED	3	700
2	2	24V_ISO	RED	3	
7	7	24V_ISO	RED	3	
8	8	24V_ISO	RED	3	
3	3	GND_ISO	BLACK	4	
4	4	GND_ISO	BLACK	4	
5	5	GND_ISO	BLACK	4	
10	10	GND_ISO	BLACK	4	
6	6	A1	BLUE	5	
12	12	K1	GREEN	6	

Note:-
1. Label the all connectors as visible at respective location

PROJECT: KAVACH		HBL HBL Engineering Ltd.	NAME		DESIGNATION	SIGN.
ASSEMBLY: EMU—MEMU			DESIGN	D.Lovaraju	Dy.Manager	
			CHECKED	J.V.Prasad	Sr.Manager	
			APPROVED	Y.Subrahmanyam	GM(Engg.)	
SCALE: NTS	SIZE: A3	TITLE: HARNS_INTRNL_RIB_CONV_EMU_MEMU HARNS ASY_24—ISO POWER SUPPLY			DRAWING NO. 516490446	REV. 1.6
	SHEET: 8 OF 13				PART NO. 6000050624	
EFFECTIVE FROM: ALL DIMENSIONS ARE IN MM			TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 — 1993. THIS DRAWING IS THE PROPERTY OF HBL ENGINEERING LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.			

REV.NO.	CHANGE DETAILS	DATE	PREPARED	APPD.
1.1	ADDED FLNG MARK& NTRL SEC WIRING	06.11.24	DLR	YS
1.2	ISOLATION SW WIRING INTERCHANGED AS PER CoE	25.06.25	DLR	YS
1.3	WIRING UPDATED WITH TRACTION CUT-OFF FEEDBACK	01.07.25	DLR	YS
1.4	WIRING LENGTHS UPDATED AS PER PRODUCTION	26.07.25	DLR	YS
1.5	BOM DETAILS ARE UPDATED AS PER PRODUCTION	05.10.25	DLR	YS
1.6	DRAWING UPDATED AS PER ICF REQUIREMENTS	29.05.26	DLR	YS



PART LIST				
ITEM NO/REF	QTY	UNITS	ITEM CODE	DESCRIPTION
1	2.00	PC	1000027692	CONNECTOR_SOCKET_12POS_MX34012SF1
2	6.00	PC	1000025249	CONNECTOR_CRIMP_TERMINAL_MX34
3	2.87	MTR	1000025384	WIRE_PTFE_RED_AWG 22_MIL-16878E/4BFB-2
4	2.87	MTR	1000013442	PTFE_WIRE_AWG 22(7/30)_BLACK
5	0.30	MTR	1000007218	FERRULE_TUBE_3.3MM_WHITE
6	0.10	MTR	1000075057	HEAT_SHRINKABLE_SLEEVE_5MM_YELLOW
7	0.15	MTR	1000075058	HEAT_SHRINKABLE_SLEEVE_5MM_CLEAR

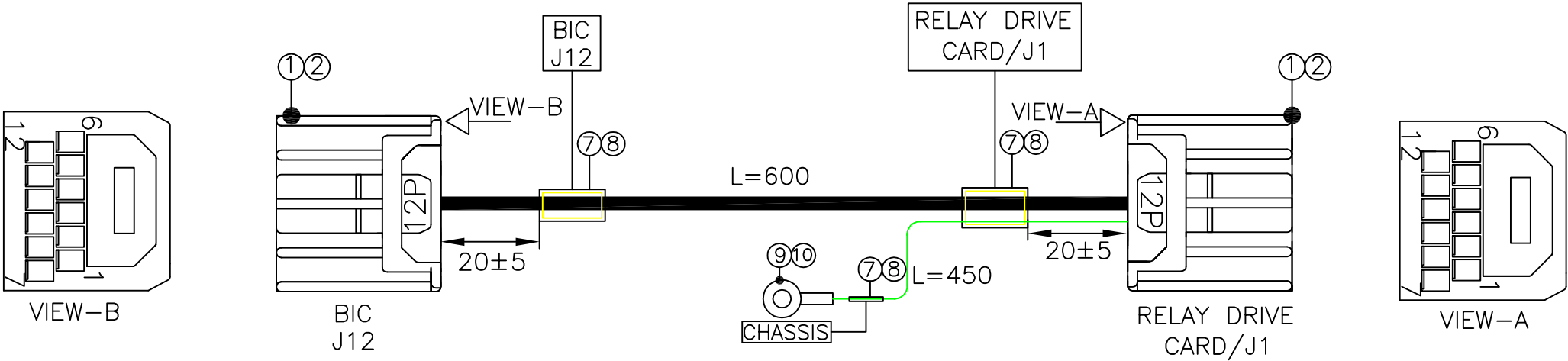
WIRE TERMINATION						
RIB LED PCB1/J1	RIB LED PCB2/J1	TO Connect.	SIGNAL	WIRE COLOUR	BOM REF#	LENGTH
2	—	RLY2/A1	CAB-A_FSB	RED	3	820
8	—	RLY2/A2		BLACK	4	
—	3	BIC J9/4	ISO_MODE	RED	3	800
—	9	ISO_RLY_5		BLACK	4	
—	4	MILB-3/K	QRV_PICKUP	RED	3	1250
—	10	MILB-3/B		BLACK	4	

Note:—

- 1.Connect the led's with respective location as per drawing
- 2.Add the ferrules to cables which are connected to relays at respective location.

PROJECT: KAVACH	HBL HBL Engineering Ltd.	NAME	DESIGNATION	SIGN.
ASSEMBLY: EMU-MEMU		DESIGN	D.Lovaraju	Dy.Manager
		CHECKED	J.V.Prasad	Sr.Manager
SCALE: NTS	SIZE: A3	APPROVED	Y.Subrahmanyam	GM(Engg.)
SHEET: 9 OF 13		TITLE: HARNS_INTRNL_RIB_CONV_EMU_MEMU HARNS ASY_RELAY LED INDICATIONS	DRAWING NO. 516490446	REV. 1.6
EFFECTIVE FROM:		PART NO. 6000050624		
ALL DIMENSIONS ARE IN MM		TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 - 1993. THIS DRAWING IS THE PROPERTY OF HBL ENGINEERING LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.		

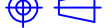
REV.NO.	CHANGE DETAILS	DATE	PREPARED	APPD.
1.1	ADDED FLNG MARK& NTRL SEC WIRING	06.11.24	DLR	YS
1.2	ISOLATION SW WIRING INTERCHANGED AS PER CoE	25.06.25	DLR	YS
1.3	WIRING UPDATED WITH TRACTION CUT-OFF FEEDBACK	01.07.25	DLR	YS
1.4	WIRING LENGTHS UPDATED AS PER PRODUCTION	26.07.25	DLR	YS
1.5	BOM DETAILS ARE UPDATED AS PER PRODUCTION	05.10.25	DLR	YS
1.6	DRAWING UPDATED AS PER ICF REQUIREMENTS	29.05.26	DLR	YS



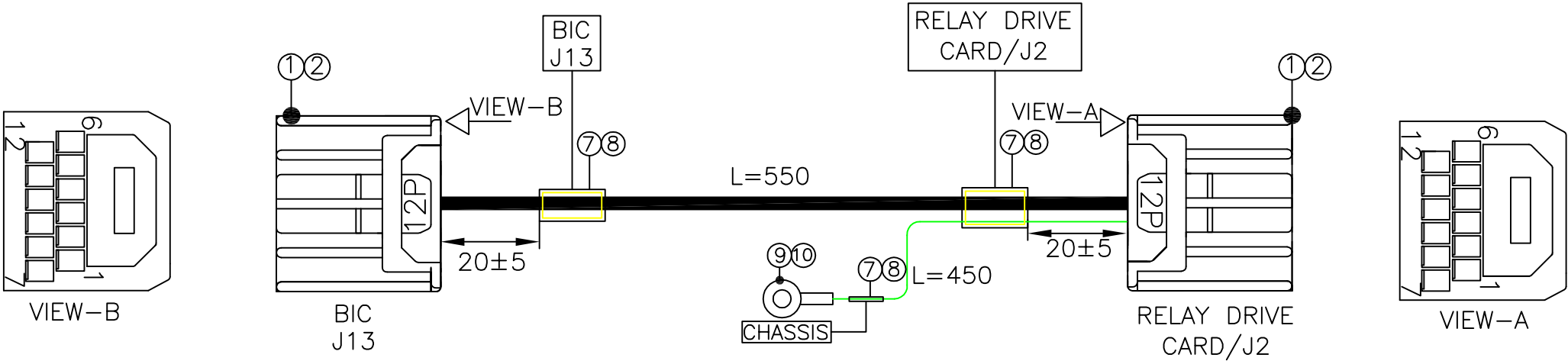
PART LIST					
ITEM NO/REF	QTY	UNITS	ITEM CODE	DESCRIPTION	
1	2.00	PC	1000027692	CONNECTOR_SOCKET_12POS_MX34012SF1	
2	12.00	PC	1000025249	CONNECTOR_CRIMP_TERMINAL_MX34	
3	1.20	MTR	1000025384	WIRE_PTFE_RED_AWG 22_MIL-16878E/4BFB-2	
4	1.20	MTR	1000013442	PTFE_WIRE_AWG 22(7/30)_BLACK	
5	0.60	MTR	1000025379	WIRE_PTFE_BLUE_AWG 22_MIL-16878E/4BFB-6	
6	0.45	MTR	1000025381	WIRE_PTFE_GREEN_AWG 22_MIL16878E/4BFB-5	
7	0.10	MTR	1000075057	HEAT_SHRINKABLE_SLEEVE_5MM_YELLOW	
8	0.15	MTR	1000075058	HEAT_SHRINKABLE_SLEEVE_5MM_CLEAR	
9	1.00	PC	1000010068	LUG_1.5SQ MM_CU_RING TYPE M6	
10	0.05	MTR	1000016883	SLEEVE_HEAT SHINKABLE_DIA 3MM_BLACK	

WIRE TERMINATION					
BIC J12	RELAY DRIVE CARD/J1	SIGNAL	WIRE COLOUR	BOM REF#	LENGTH
1	1	24V_ISO	RED	3	600
2	2	GND_ISO	BLACK	4	
7	7	24V_ISO	RED	3	
8	8	GND_ISO	BLACK	4	
4	4	FOULING_MARK	BLUE	5	
—	6	CHASSIS	GREEN	6	450

Note:—
1. Label the all connectors as per respective location.

PROJECT: KAVACH		HBL HBL Engineering Ltd.	NAME		DESIGNATION	SIGN.
ASSEMBLY: EMU—MEMU			DESIGN	D.Lovaraju	Dy.Manager	
			CHECKED	J.V.Prasad	Sr.Manager	
			APPROVED	Y.Subrahmanyam	GM(Engg.)	
SCALE: NTS	SIZE: A3	TITLE: HARNS_INTRNL_RIB_CONV_EMU_MEMU HARNS ASY_FOULING MARK INPUT			DRAWING NO. 516490446	REV. 1.6
	SHEET: 10 OF 13				PART NO. 6000050624	
EFFECTIVE FROM: ALL DIMENSIONS ARE IN MM			TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 – 1993. THIS DRAWING IS THE PROPERTY OF HBL ENGINEERING LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.			

REV.NO.	CHANGE DETAILS	DATE	PREPARED	APPD.
1.1	ADDED FLNG MARK& NTRL SEC WIRING	06.11.24	DLR	YS
1.2	ISOLATION SW WIRING INTERCHANGED AS PER CoE	25.06.25	DLR	YS
1.3	WIRING UPDATED WITH TRACTION CUT-OFF FEEDBACK	01.07.25	DLR	YS
1.4	WIRING LENGTHS UPDATED AS PER PRODUCTION	26.07.25	DLR	YS
1.5	BOM DETAILS ARE UPDATED AS PER PRODUCTION	05.10.25	DLR	YS
1.6	DRAWING UPDATED AS PER ICF REQUIREMENTS	29.05.26	DLR	YS



PART LIST				
ITEM NO/REF	QTY	UNITS	ITEM CODE	DESCRIPTION
1	2.00	PC	1000027692	CONNECTOR_SOCKET_12POS_MX34012SF1
2	12.00	PC	1000025249	CONNECTOR_CRIMP_TERMINAL_MX34
3	1.10	MTR	1000025384	WIRE_PTFE_RED_AWG 22_MIL-16878E/4BFB-2
4	1.10	MTR	1000013442	PTFE_WIRE_AWG 22(7/30)_BLACK
5	0.55	MTR	1000025379	WIRE_PTFE_BLUE_AWG 22_MIL-16878E/4BFB-6
6	0.45	MTR	1000025381	WIRE_PTFE_GREEN_AWG 22_MIL16878E/4BFB-5
7	0.10	MTR	1000075057	HEAT_SHRINKABLE_SLEEVE_5MM_YELLOW
8	0.15	MTR	1000075058	HEAT_SHRINKABLE_SLEEVE_5MM_CLEAR
9	1.00	PC	1000010068	LUG_1.5SQ MM_CU_RING TYPE M6
10	0.05	MTR	1000016883	SLEEVE_HEAT SHINKABLE_DIA 3MM_BLACK

WIRE TERMINATION					
BIC J13	RELAY DRIVE CARD/J2	SIGNAL	WIRE COLOUR	BOM REF#	LENGTH
1	1	24V_ISO	RED	3	550
2	2	GND_ISO	BLACK	4	
7	7	24V_ISO	RED	3	
8	8	GND_ISO	BLACK	4	
4	4	NEUTRAL_SEC	BLUE	5	
-	6	CHASSIS	GREEN	6	450

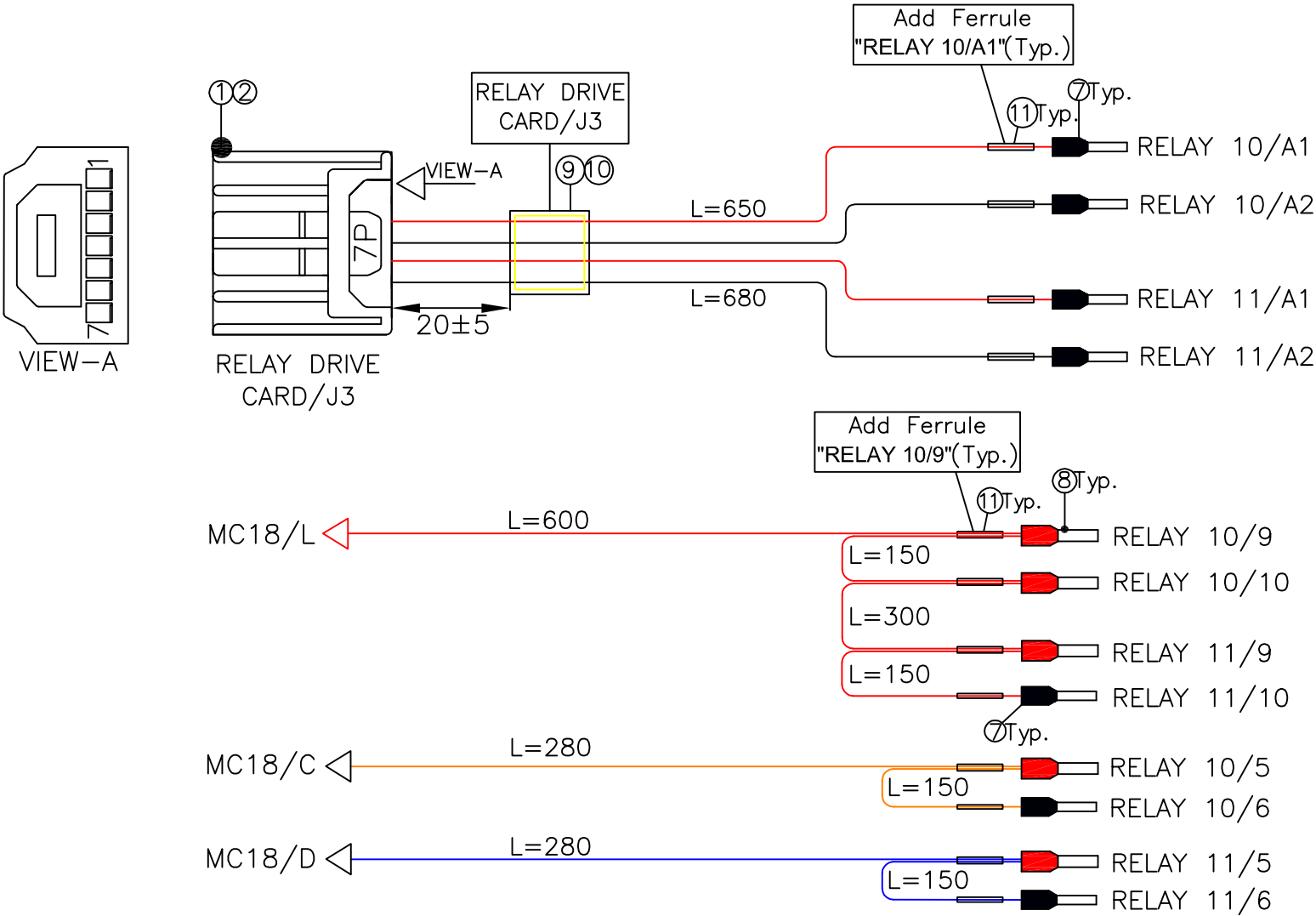
Note:-
1. Label the all connectors as per respective location.

PROJECT: KAVACH		HBL HBL Engineering Ltd.	NAME	DESIGNATION	SIGN.
ASSEMBLY: EMU—MEMU			DESIGN	D.Lovaraju	Dy.Manager
			CHECKED	J.V.Prasad	Sr.Manager
			APPROVED	Y.Subrahmanyam	GM(Engg.)
SCALE: NTS	SIZE: A3	TITLE: HARNS_INTRNL_RIB_CONV_EMU_MEMU HARNS ASY_NEUTRAL SECTION INPUT	DRAWING NO. 516490446		REV. 1.6
	SHEET: 11 OF 13		PART NO. 6000050624		
EFFECTIVE FROM: ALL DIMENSIONS ARE IN MM			TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 – 1993. THIS DRAWING IS THE PROPERTY OF HBL ENGINEERING LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.		

REV.NO.	CHANGE DETAILS	DATE	PREPARED	APPD.
1.1	ADDED FLNG MARK& NTRL SEC WIRING	06.11.24	DLR	YS
1.2	ISOLATION SW WIRING INTERCHANGED AS PER CoE	25.06.25	DLR	YS
1.3	WIRING UPDATED WITH TRACTION CUT-OFF FEEDBACK	01.07.25	DLR	YS
1.4	WIRING LENGTHS UPDATED AS PER PRODUCTION	26.07.25	DLR	YS
1.5	BOM DETAILS ARE UPDATED AS PER PRODUCTION	05.10.25	DLR	YS
1.6	DRAWING UPDATED AS PER ICF REQUIREMENTS	29.05.26	DLR	YS

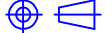
WIRE TERMINATION-1						
RELAY DRIVE CARD/J3	RELAY10	RELAY11	SIGNAL	WIRE COLOUR	BOM REF#	LENGTH
1	A1	—	24V	RED	4	650
3	A2	—	FLNG_MRK_COIL	BLACK	3	
5	—	A1	24V	RED	4	680
7		A2	NTRL_SEC_COIL	BLACK	3	

WIRE TERMINATION-2						
MC18	RLY10	RLY11	SIGNAL	WIRE COLOUR	BOM REF#	LENGTH
C	5&6	—	FOULING_MARK	ORANGE	5	280+150
D	—	5&6	NUETRAL_SECTION	BLUE	6	280+150
L	9&10	9&10	LOCO_+VE	RED	4	1200



PART LIST				
ITEM NO/REF	QTY	UNITS	ITEM CODE	DESCRIPTION
1	1.00	PC	1000041208	CONNECTOR_SOCKET_7POS_MX34007SF1
2	4.00	PC	1000025249	CONNECTOR_CRIMP_TERMINAL_MX34
3	1.33	MTR	1000013442	PTFE_WIRE_AWG 22(7/30)_BLACK
4	2.53	MTR	1000025384	WIRE_PTFE_RED_AWG 22_MIL-16878E/4BFB-2
5	0.43	MTR	1000025383	WIRE_PTFE_ORANGE_AWG22_MIL-16878E/4BFB-3
6	0.43	MTR	1000025379	WIRE_PTFE_BLUE_AWG 22_MIL-16878E/4BFB-6
7	7.00	PC	1000095964	FERRULE_INSULATED_0.5MM_58-508L8_JAINSON
8	5.00	PC	1000095965	FERRULE_INSULATED_1MM_58-510L8_JAINSON
9	0.05	MTR	1000075057	HEAT_SHRINKABLE_SLEEVE_5MM_YELLOW
10	0.06	MTR	1000075058	HEAT_SHRINKABLE_SLEEVE_5MM_CLEAR
11	0.20	MTR	1000007218	FERRULE_TUBE_3.3MM_WHITE

Note:-
1. Label the all connectors as per respective location.

PROJECT: KAVACH		HBL HBL Engineering Ltd.		NAME	DESIGNATION	SIGN.
ASSEMBLY: EMU—MEMU			DESIGN	D.Lovaraju	Dy.Manager	
			CHECKED	J.V.Prasad	Sr.Manager	
			APPROVED	Y.Subrahmanyam	GM(Engg.)	
SCALE: NTS	SIZE: A3	TITLE: HARNS_INTRNL_RIB_CONV_EMU_MEMU HARNS ASY_FLNG MARK&NTRL SECTION	DRAWING NO. 516490446			REV. 1.6
	SHEET: 12 OF 13		PART NO. 6000050624			
EFFECTIVE FROM: ALL DIMENSIONS ARE IN MM		TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 – 1993. THIS DRAWING IS THE PROPERTY OF HBL ENGINEERING LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.				

REV.NO.	CHANGE DETAILS	DATE	PREPARED	APPD.
1.1	ADDED FLNG MARK& NTRL SEC WIRING	06.11.24	DLR	YS
1.2	ISOLATION SW WIRING INTERCHANGED AS PER CoE	25.06.25	DLR	YS
1.3	WIRING UPDATED WITH TRACTION CUT-OFF FEEDBACK	01.07.25	DLR	YS
1.4	WIRING LENGTHS UPDATED AS PER PRODUCTION	26.07.25	DLR	YS
1.5	BOM DETAILS ARE UPDATED AS PER PRODUCTION	05.10.25	DLR	YS
1.6	DRAWING UPDATED AS PER ICF REQUIREMENTS	29.05.26	DLR	YS

PART LIST						
ITEM NO/REF	QTY	UNITS	ITEM CODE	DESCRIPTION	MANUFACTURER	PART NO.
1	1.00	PC	1000025258	CONNECTOR_SOCKET_24POS_PNO:MX34024SF1	JAE CONNECTORS	MX34024SF1
2	2.00	PC	1000025257	CONNECTOR_SOCKET_20POS_PNO:MX34020SF1	JAE CONNECTORS	MX34020SF1
3	2.00	PC	1000025256	CONNECTOR_SOCKET_16POS_PNO:MX34016SF1	JAE CONNECTORS	MX34016SF1
4	8.00	PC	1000027692	CONNECTOR_SOCKET_12POS_MX34012SF1	JAE CONNECTORS	MX34012SF1
5	2.00	PC	1000041208	CONNECTOR_SOCKET_7POS_MX34007SF1	JAE CONNECTORS	MX34007SF1
6	2.00	PC	1000027695	CONNECTOR_SOCKET_5POS_MX34005SF1	JAE CONNECTORS	MX34005SF1
7	112.00	PC	1000025249	CONNECTOR_CRIMP_TERMINAL_MX34	JAE CONNECTORS	M34S75C4F2
8	1.00	PC	1000050114	CONNECTOR_PLUG_19PIN_62IN-5024-14-19P	AMPHENOL	62IN50241419P
9	1.00	PC	1000088342	CONNECTOR_RECEP_14P_62IN-5024-14-A14-P	AMPHENOL	62IN502414A14P
10	1.00	PC	1000088343	CONNECTOR_RECEP_14P_62IN-5024-14-A14-S	AMPHENOL	62IN502414A14S
11	1.00	PC	1000088346	CONNECTOR_RECEP_12P_62IN-5024-14-12P	AMPHENOL	62IN50241412P
12	2.00	PC	1000088347	CONNECTOR_RECEP_12P_62IN-5024-14-12S	AMPHENOL	62IN50241412S
13	2.00	MTR	1000016883	SLEEVE_HEAT SHINKABLE_DIA 3MM_BLACK	TYCO	-
14	19.00	MTR	1000013442	PTFE_WIRE_AWG 22(7/30)_BLACK	Tonk & Associates	-
15	2.20	MTR	1000025380	WIRE_PTFE_BROWN_AWG 22_MIL-16878E/4BFB-1	Tonk & Associates	-
16	20.50	MTR	1000025384	WIRE_PTFE_RED_AWG 22_MIL-16878E/4BFB-2	Tonk & Associates	-
17	3.80	MTR	1000025383	WIRE_PTFE_ORANGE_AWG22_MIL-16878E/4BFB-3	Tonk & Associates	-
18	8.00	MTR	1000025386	WIRE_PTFE_YELLOW_AWG 22_MIL16878E/4BFB-4	Tonk & Associates	-
19	4.50	MTR	1000025381	WIRE_PTFE_GREEN_AWG 22_MIL16878E/4BFB-5	Tonk & Associates	-
20	7.50	MTR	1000025379	WIRE_PTFE_BLUE_AWG 22_MIL-16878E/4BFB-6	Tonk & Associates	-
21	0.70	MTR	1000025385	WIRE_PTFE_VIOLET_AWG22_MIL-16878E/4BFB-7	Tonk & Associates	-
22	4.80	MTR	1000025382	WIRE_PTFE_GREY_AWG 22_MIL-16878E/4BFB-8	Tonk & Associates	-
23	1.50	MTR	1000026453	PTFE_WIRE_AWG 22(7/30)_WHITE	Tonk & Associates	-
24	3.00	PC	1000010068	LUG_1.5SQ MM_CU_RING TYPE M6	Reputed	-
25	10.00	PC	1000095965	FERRULE_INSULATED_1MM_58-510L8_JAINSON	Reputed	-
26	21.00	PC	1000095964	FERRULE_INSULATED_0.5MM_58-508L8_JAINSON	Reputed	-
27	1.00	PC	1000014018	RESISTOR_MFR_47E_1/4W_1%	Reputed	-
28	2.00	PC	1000075399	DIODE_SCHOTTKY_LEADED_40V_5A	STMicroelectronics	STPS5L40RL
29	1.00	PC	1000003441	CAPACITOR_MPEAR(AXIAL)_0.47MFD/250V	Reputed	-
30	0.30	MTR	1000026054	SLEEVE_HEAT SHRINKABLE_5MM_BLACK	Reputed	-
31	1.00	MTR	1000075057	HEAT_SHRINKABLE_SLEEVE_5MM_YELLOW	WOER	5053164039
32	1.50	MTR	1000075058	HEAT_SHRINKABLE_SLEEVE_5MM_CLEAR	WOER	5070074032
33	0.50	MTR	1000074458	HEAT-SHRINKABLE_SLEEVE_16MM_BLACK	WOER	-
34	2.00	MTR	1000007218	FERRULE TUBE_3.3MM_WHITE	WOER	-
35	3.00	PC	1000006449	DIODE_LEADED_1N4007_DO-41	-	-
36	1.00	PC	1000024955	DIODE_LEADED_1.5KE36CA_36V/1500W	-	-
37	100.00	PC	1000001886	CABLE TIES FR TYPE_100MM	-	-
38	0.30	MTR	1000016947	SLEEVE_MARKER_3/8"_YELLOW	-	-
39	0.40	MTR	1000016866	SLEEVE_CLEAR_3/8"	-	-

PROJECT: KAVACH		HBL HBL Engineering Ltd.		NAME	DESIGNATION	SIGN.
ASSEMBLY: EMU—MEMU			DESIGN	D.Lovaraju	Dy.Manager	
			CHECKED	J.V.Prasad	Sr.Manager	
			APPROVED	Y.Subrahmanyam	GM(Engg.)	
SCALE: NTS	SIZE: A3	TITLE: HARNS_INTRNL_RIB_CONV_EMU_MEMU BOM SUMMARY SHEET			DRAWING NO. 516490446	REV. 1.6
	SHEET: 13 OF 13				PART NO. 6000050624	
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